

Bayesian Hidden Markov Models to Infer Latent Pitcher States

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1. Motivation

What Problem Does This Method Solve?

Why Can't Standard Regression Handle It?

Where Does It Show Up in Practice?

2. Model Specification

Likelihood

Priors

Hierarchical Structure

Key Modeling Assumptions

3. Applied Example

Data Description and Context

Load Data

Fit Model

MCMC Diagnostics

Posterior Summaries

Posterior Predictive Checks

Visualizations

4. Interpretation

What Did We Learn?

Limitations

When Should We Use This Model?

Sensitivity to Modeling Assumptions